

BANARAS HINDU UNIVERSITYB.Sc. (Hons.) — Semester I Mid-Semester Examination, 2024-25 (NEP)

PHYSICS (SEC)**Paper Code: PHYSE-11 : Mathematical Problem Solving Skills (Mid-Sem)**

Time: 1 Hour

Maximum Marks: 20

Instructions: Answer all questions. Each question carries equal marks (5 marks each).

Question 1. Use Taylor theorem to evaluate $\sqrt{1.5}$. Please limit yourself to using only the first 4 terms of the Taylor expansion. [5]

Question 2. Draw the graph of function $f(x) = x^2 - 1$. Find the slope of the tangent line at the points $(-1, 0)$, $(0, -1)$, and $(2, 3)$. [5]

Question 3. What is the difference between ordinary and partial differential equations? Explain with examples. [5]

Question 4. Solve the following differential equation by using the integrating factor:

$$x \frac{dy}{dx} + 2y = 10x^2$$

[5]